

Lecture 7 9/15/22 more contradiction

Want to prove P .

Assume $\neg P$.

Argue.

Conclude Q .

Observe that Q is false

Conclude P .

$$\neg P \Rightarrow Q$$

Prove that $(\forall n \in \mathbb{Z}, n \text{ and } n+1$
have no common prime factors)

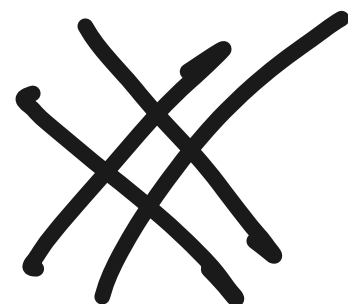
Proof: Proceed by contradiction. Assume
 $\exists n \in \mathbb{Z}$ s.t. n and $n+1$ have some
common prime divisors. Let p be a prime
s.t. $p|n$ and $p|n+1$.

Then $p \mid (n+1) - n$, i.e., $p \mid 1$.

This is a contradiction, since

p is prime, and primes are > 1 .

$\Rightarrow \Leftarrow$



$$d \mid a \wedge d \mid b \Rightarrow d \mid a+b$$

$$d \mid a \Rightarrow d \mid -a$$

$$d \mid a \wedge b \quad d \mid a-b$$

Let $a, b, c \in \mathbb{Z}$. Sps $a^2 + b^2 = c^2$.

Hyp

Show that abc is even.

Proof: Proceed by contradiction.

Assume $a^2 + b^2 = c^2$ and abc is odd.

Then a, b , and c are each odd. (If it were even, abc would be even)

Then $a^2, b^2, +c^2$ are odd (b/c products of odd integers are odd). Then

$a^2 + b^2$ is even, but c^2 is odd.

This is "even = odd" which

is a contradiction. $\square$

Defn: A number x is rational if
 $\exists a, b \in \mathbb{Z}$ s.t. $b \neq 0$ and $x = a/b$.

A rational # is reduced if a and b
have no common prime divisors

Examples: $\frac{2}{3}$ = $\frac{4}{6}$
reduced not reduced.

Fact: Every rational #
can be reduced

Prove: $\sqrt{2} \notin \mathbb{Q}$. (It $\sqrt{2}$ is not Rational.)

Proof: Proceed by contradiction Assume $\sqrt{2} \in \mathbb{Q}$.

Then $\exists a, b \in \mathbb{Z}$ s.t. $b \neq 0$ and $\sqrt{2} = a/b$.

Assume that a & b are reduced. In particular at least one of a or b is odd. Then

$$b\sqrt{2} = a. \text{ Then } b^2 \cdot 2 = a^2.$$

Since the LHS is even, the RHS is even, i.e.,
 a^2 is even. Thus a is even. (By HW1, if
 a is odd a^2 is odd.) Then $4 \mid a^2$; indeed,
 if a is even then $a = 2k$ for some $k \in \mathbb{Z}$. Then $b^2 \cdot 2 = (2k)^2 = 4k^2$.
 Then $b^2 = 2k^2$. Since the RHS is even, b^2 is even,
 so b is even. This is a contradiction,

Since $a+b$ are not both odd.

HW: $\sqrt{3} \notin \mathbb{Q}$

$2/3$
 3

tan 1

$\pi = 3.141\dots$

e

Prove: $\log_3 2 \notin \mathbb{Q}$.

Recall: $3^{\log_3 2} = 2$

Proof: Proceed by contradiction.

Assume $\log_3 2 \in \mathbb{Q}$. Then $\exists a, b \in \mathbb{Z}$
s.t. $b \neq 0$ and $\log_3 2 = \frac{a}{b}$. w.m.a

(we may assume) a & b are reduced.

$$\Rightarrow 2 = 3^{\log_3 2} = 3^{\frac{a}{b}}.$$

Then $2^b = 3^a$. $(3^{a/b})^b = 3^a$

Since $b > 0$, the LHS is even.

But the RHS is odd. This is

a contradiction. $\square$

Prove: if $a \in \mathbb{Q}$ and $b \notin \mathbb{Q}$, then
 $a+b \notin \mathbb{Q}$.

Proof: Assume $a \in \mathbb{Q}$ and $b \notin \mathbb{Q}$. Proceed by
contradiction. Assume $a+b \in \mathbb{Q}$. Then
 $\exists c, d \in \mathbb{Z}$ s.t. $d \neq 0$ and $a = c/d$. Then
 $\exists e, f \in \mathbb{Z}$ s.t. $f \neq 0$ and $a+b = e/f$

$$\text{Then } b = (a+b) - a = \frac{e}{f} - \frac{c}{d} = \frac{ed - cf}{fd}.$$

Since $ed - cf, fd \in \mathbb{Z}$ and $fd \neq 0$,

$b \in \mathbb{Q}$. This contradicts $b \notin \mathbb{Q}$.

We conclude that $a+b \notin \mathbb{Q}$. □

Let a, b, c be odd. Let x be a solution to $ax^2 + bx + c = 0$. Prove $x \notin \mathbb{Q}$.

Proof, Assume a, b, c are odd.

Assume $ax^2 + bx + c = 0$.

Proceed by contradiction. Assume $x \in \mathbb{Q}$.

Then $\exists d, e \in \mathbb{Z}$ s.t. $e \neq 0$ and
 $x = d/e$. WMA d & e are reduced.

Then $a\left(\frac{d}{e}\right)^n + b\frac{d}{e} + c = 0$. \begin{equation}

Then $a d^n + b d e + c e^2 = 0$. (1.1)

Since d & e are reduced, at least one

is odd.

There are 3 cases: d & e are odd, d is even & e is odd, or d is odd & e is even

In case 1, the LHS of (1.1) is "odd field + odd" = odd = even, a contradiction.

In case 2, the LHS is " $e + e + 1 = e$ ", ~~9~~

Case 3 is similar.

In each case, we have a contradiction 

$2, 3, 5, 7, 11 \dots$ $p \neq 1$ and
 p is prime if $p = ab \implies$
 $a = \pm 1$ or $b = \pm 1$

$\forall a, b \text{ s.t. } p = ab,$
 $a = \pm 1$ or $b = \pm 1$

Fermat: $1 = 1^2 = 1^2 + 0^2 = \dots$

Fermat:

$$2^0 + 1 = 1 + 1 = 2$$

$$2^1 + 1 = 2 + 1 = 3$$

$$2^2 + 1 = 4 + 1 = 5$$

$$2^4 + 1 = 16 + 1 = 17$$

$$2^8 + 1 = 256 + 1 = 257$$

$$2^{16} + 1 = 65536 + 1 = 65537$$

Fermat conjrd that $\forall n, 2^{2^n} + 1$ is

False prime.

Modern conjrd $2^{2^n} + 1 \nexists$ new prime
if $n \geq 5$.

$$2^3 + 1 = 8 + 1 = 9 = 3 \cdot 3$$

$$2^5 + 1 = 32 + 1 = 33 = 3 \cdot 11$$

$$2^7 + 1 = 128 + 1 = \underline{129} = 3 \cdot 43$$

Problem: if $2^n + 1$ is prime then n is
even.

Proof: Assume $2^n + 1$ is prime.

Proceed by contradiction. Assume n is

odd.
$$\left(\begin{aligned} n^2 - 1 &= (n-1)(n+1) \\ x^n - y^n &= (x-y)(x^{n-1} + x^{n-2}y + \dots + y^{n-1}) \end{aligned} \right)$$

Since n is odd, $(-1)^n = -1$.

$$\text{Thus } 2^n + 1 = 2^n + (-1)^n = 2^n - (-1)^n.$$

$$\text{This factors as } (2 - (-1)) (2^{n-1} - 2^{n-2} + \dots \pm 1) \\ = 3 \cdot ?$$

If $2^n + 1 > 3$, it is not prime b/c
3 is a proper divisor. $\square$